

CORE JAVA

```
class Mythread extends Thread{

    public void run(){
        try {

            // Thread.sleep(10000);
        } catch (Exception e) {
            // TODO: handle exception
        }
        System.out.println("Run method using Thread class");
        System.out.println(Thread.currentThread().getName());
    }

    public void nonRUN(){
        System.out.println("non RUN method ");
    }
}

// interface Demo{

//     public void run();
//     public void inter();
// }

class MyRunnable implements Runnable{

    public void run(){
        System.out.println(Thread.currentThread().getName());
        System.out.println("Run method inside Runnable interface");
    }
}

public class Multithreading {
    public static void main(String[] args) throws InterruptedException{

        MyRunnable obj1 = new MyRunnable();

        Thread obj2 = new Thread(obj1);

        obj2.start();
        obj2.setName("Runnable class");
    }
}
```

```

    Mythread obj = new Mythread();

    obj.start();
    obj.setName("Thread class");

    // obj.join();
    obj.nonRun();

    Thread.sleep(20000);

    // two ways => thread class , runnable interface
    System.out.println("Multithreading");
}
}

```

```

class Counter{

    int count = 0;

    synchronized void increment(){
        count++;
    }
}

public class SynchronizeThread {

    public static void main(String[] args) throws InterruptedException{

        Counter cnt = new Counter();

        Thread obj = new Thread( ()->{
            for(int i=1; i<=25000; i++){
                cnt.increment();
            }
        } );

        Thread obj1 = new Thread( ()->{
            for(int i=1; i<=25000; i++){
                cnt.increment();
            }
        } );
    }
}

```

```

        obj.start();
        obj1.start();

        obj.join();
        obj1.join();

        System.out.println(cnt.count);
        System.out.println("Main thread");
    }
}

```

```

import java.util.ArrayList;

public class Collections {

    public static void main(String[] args) {
        System.out.println("Collections");

        ArrayList<String> lt = new ArrayList<>();

        lt.add("FIIT");
        lt.add("FIIT");
        lt.add("1");
        lt.add("2");
        lt.add("3");

        lt.remove("FIIT");

        lt.set(3, "Academy");
        System.out.println(lt.contains("dfasd"));

        System.out.println(lt);

        System.out.println(lt.get(0));

        System.out.println(lt.size());

        // ArrayList<Integer> ltt = new ArrayList<>();

        // wrapper class => primitive to non primitive

        // list interface
    }
}

```

```
        // allow duplicates
        // maintains insertion order

    // arraylist
        // manipulation is slow
        // accessing element is fast (because index)
    }
}
```